3-State Empirical Transition Tensor Calculation Complete

Your query successfully calculated the empirical payout distribution conditioned on **machine_type**, **signal**, and the previous two payouts (**lag1** and **lag2**), with Laplace smoothing (α = 1.0) applied to all probability estimates.

Key Results

The analysis produced **259 unique state combinations** across all training/validation ticks (before cutoff), covering:

• **3 machine types**: balanced, jackpot, steady
• **3 signal values**: -1, 0, 1
• **6 lag1 values**: -1 (initial state), 0, 1, 2, 5, 20
• **6 lag2 values**: -1 (initial state), 0, 1, 2, 5, 20

Each row contains:

• **Raw counts** for each payout outcome (0, 1, 2, 5, 20)
• **Total samples** observed in that state
• **Laplace-smoothed probabilities** (p_0, p_1, p_2, p_5, p_20) calculated as: (count + 1) / (total + 5)

Sample Insights

High-frequency states (e.g., balanced machines with signal=1, lag1=0, lag2=0) have **7,077 samples** with strong empirical evidence showing ~70% probability of payout=0.

Low-frequency states (e.g., steady machines with signal=1, lag1=20, lag2=5) have only **1 sample**, where Laplace smoothing prevents zero probabilities by assigning ~0.167 to each outcome.

The tensor is now ready for prediction tasks—simply look up the (machine_type, signal, lag1, lag2) state and use the corresponding probability distribution for the next payout.